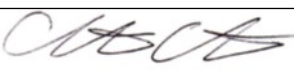


**Guideline Title: Pediatric Ventricular Fibrillation –
Pulseless Ventricular Tachycardia**

Guideline Number: GUID-3203-PED013

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Ensure prompt identification and provide intervention to restore a stable cardiac rhythm with adequate cardiac output and.
- II. **PATHOPHYSIOLOGY:** VF/VT is uncommon in the pediatric age group, and the patients typically have underlying structural heart disease. Other causes include Brugada (a genetic disease), and prolonged QT syndromes, hypoxia, acidosis, electrolyte imbalance, hypothermia, or poisoning.
- III. **DEFINITIONS:**
 - A. VF – Ventricular Fibrillation
 - B. VT – Ventricular Tachycardia
- IV. **DEPARTMENT GUIDELINE:**
 - A. Initiate CPR per AHA guideline with compression rate of 100-120/min.
 - B. Apply cardiac monitor and confirm VF/VT. Place defibrillation pads on patient.
 - C. Defibrillation at 2 J/kg. Immediately resume CPR for 5 cycles or approximately 2 minutes.
 - i. Repeat defibrillation at 4 J/kg for subsequent shock. For additional defibrillations, increase by 2J/kg up to a maximum of 10J/kg (or adult dose) noting the maximum allowable Joule setting per manufacture’s guidelines.
 - ii. Perform rhythm analysis every 2 minutes between cycles of CPR.
 - D. Obtain or confirm IV/IO access.
 - E. Manage airway and ensure adequate oxygenation and ventilation with positive pressure ventilation.
 - i. Initially administer O2 via BVM
 - ii. Secure airway and ventilate with 100% oxygen. Intubate, if necessary, per *GUID3203-PR020 Endotracheal Intubation*.
 - iii. Ventilate with 1 breath every 2 to 3 seconds (a rate of about 20-30 breaths per minute). Avoid hyperventilation.
 - iv. Apply ETCO2 monitor if advanced airway present.
 - F. Administer
 - i. **Epinephrine (1:10,000) 0.01 mg/kg (0.1 ml/kg) IV/IO**, maximum dose 1 mg.
 - ii. **Epinephrine (1:1,000) 0.1 mg/kg (0.1 ml/kg) via ETT**, max dose 1 mg
 - iii. Re-dose every 3-5 minutes
 - G. Administer an anti-arrhythmic

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- i. **Amiodarone 5 mg/kg IV/IO Bolus, maximum dose 300 mg.** May repeat bolus up to 2 times for a maximum daily dose of 15mg/kg for refractory VF/pulseless VT.
 - ii. **Lidocaine 1 mg/kg IV/IO, maximum dose 100 mg** repeated every 5 minutes as needed at 0.5 mg/kg IV/IO, until the dysrhythmia is suppressed or 3 mg/kg have been given. If the dysrhythmia is suppressed, follow with a 20 - 50 mcg/kg/min infusion.
 - iii. For polymorphic VT or torsade's de pointes administer **Magnesium sulfate 50 mg/kg IV/IO**
- H. For deterioration during transport after ROSC, consider mechanical problems with the airway, ventilation, and/or blood volume. Dysrhythmias are rare, except as an agonal event. Consider: extubation, plugged endotracheal tube, right main stem bronchus intubation, distended stomach causing impaired lung expansion, pneumothorax, inadequate minute ventilation, ventilator malfunction, disconnected or depleted oxygen supply, hypovolemic shock, or agitation.
- I. Report Broselow color and/or estimated weight to receiving center prior to arrival
- J. Out of hospital cessation of resuscitation efforts should not be considered in pediatric patients without contacting medical control.
- K. Always consider the latest recommendations from the American Heart Association
- V. **DOCUMENTATION:**
 - A. EMS Charts
- VI. **REFERENCES:**
 - A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.